
ME 639: Thermal Power Plant Engineering

Course Code	ME 639
Course Name	Thermal Power Plant Engineering
L-T-P-C	3-0-0-3
Prerequisites	Instructor Consent
Intended for	UG/M. Tech. /MS/PhD
Distribution	Specialized stream elective course for M. Tech. in Mechanical Engineering with specialization in Energy Systems, and elective course for other students
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I**

Types of thermal power stations, steam power stations based on fossil fuels, Economy and thermal scheme of the steam power stations. [4 Lectures]

- **Module II**

Thermal power plant equipment – boilers, superheaters, super critical steam generator economizers, feed-water heaters, condensers. [20 Lectures]

- **Module III**

Combustion chamber and gas loop, turbines, cooling towers, etc. Gas turbine power stations, combined cycle power plants. [12 Lectures]

- **Module IV**

Internal combustion engine plant for peak load, standby and start-up. [6 Lectures]

Text Books

1. Amiya Ranjan Mallick, **Practical Boiler Operation Engineering and Power Plant**, 4th Edition, PHI, 2015.
2. R. Yadav, **Steam and Gas Turbines and Power Plant Engineering**, 7th Edition, Central Publishing House, 2000.